

Shitesh Keshav Khaw

Full-Stack AI Engineer | Next.js | FastAPI | React | RAG | Cloud Systems

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LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Full-Stack AI Engineer with experience building production-grade SaaS platforms, scalable backend systems, cloud-native applications, and advanced AI-powered products. Skilled in designing multi-tenant architectures, REST APIs, authentication systems, database models, and Retrieval-Augmented Generation (RAG) workflows using Next.js, FastAPI, PostgreSQL, React, and AWS. Experienced in owning products end-to-end, from architecture and infrastructure to deployment, observability, and user-facing experiences.

TECHNICAL EXPERTISE

Languages: JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js, Tailwind CSS, Framer Motion, Zustand, React Query, SWR

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, OAuth2, NextAuth, Zod, Pydantic

Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM, ChromaDB

AI Engineering: Retrieval-Augmented Generation (RAG), Vector Search, Cross-Encoder Reranking, Semantic Search, Prompt Engineering, OpenAI API, OpenRouter, Groq API, LLaMA 3, Generative AI Systems

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Docker Compose, Oracle Cloud, Vercel, CI/CD, GitHub Actions, Nginx

Architecture & Systems: Multi-Tenant SaaS, RBAC, Audit Logging, Event-Driven Architecture, Distributed Systems, API Design, System Design

PROFESSIONAL EXPERIENCE

Lead AI Systems Architect — Makhija Quantum AI

Nov 2025 – Present

Mumbai, India

- Architected and deployed a highly available AI automation infrastructure, decoupling backend services from workflow orchestration to ensure scalable fault tolerance for B2B client onboarding.
- Engineered a high-throughput backend service layer using Python, eliminating manual lead-routing bottlenecks by automating 100% of incoming multi-channel communication workflows.
- Provisioned PostgreSQL databases on AWS RDS with a multi-tenant schema architecture, prioritizing strict data isolation and scalable ACID compliance.
- Designed a robust CI/CD strategy utilizing Docker across AWS and Oracle Cloud, maintaining continuous delivery and minimizing production downtime.
- Built complex, event-driven n8n automation pipelines processing multi-step data transformations across WhatsApp, Instagram, and LinkedIn webhooks with zero-downtime event logging.

KEY ARCHITECTURAL PROJECTS

DocuMindAI — An Enterprise Document Intelligence Platform with Retrieval, Validation, Compliance, Contradiction Analysis, and Explainable AI.

Live Platform — [GitHub Repository](#)

- Architected and developed a production-grade AI platform enabling organizations to analyze, validate, and interact with large-scale unstructured information through conversational and analytical workflows.
- Engineered a hybrid Retrieval-Augmented Generation (RAG) pipeline combining vector search and Cross-Encoder reranking, improving retrieval precision and reducing irrelevant context during LLM inference.
- Designed a contradiction intelligence engine capable of identifying conflicting facts, dates, monetary values, requirements, and semantic inconsistencies across multiple information sources.
- Built explainable AI workflows including citation grounding, retrieval diagnostics, trust scoring, and traceability systems to increase transparency and reduce hallucinations.
- Implemented secure multi-tenant architecture with RBAC authorization, workspace isolation, JWT authentication, and immutable audit logging.

- Developed scalable asynchronous processing pipelines for ingestion, indexing, retrieval, validation, and AI-powered analysis of large document collections.

TaskOS — AI-Powered Productivity Operating System

Live Platform — [GitHub Repository](#)

- Architected a domain-driven Next.js 15.5 application utilizing a Neon Serverless PostgreSQL database, engineering 12 heavily indexed relational models with Prisma ORM for type-safe data transactions.
- Integrated LLaMA 3 70B via the Groq API to power a real-time AI productivity coach, implementing specialized domain layers for algorithmic task decomposition and streak-risk analysis.
- Engineered a robust authentication pipeline combining Google OAuth 2.0 with custom JWT middleware guards, enforcing rate limiting and Zod-based strict request validation across 14 API route groups.
- Designed a decoupled micro-service architecture to handle complex asynchronous tasks, including Web Push (VAPID) notifications, AWS S3 presigned avatar uploads, and Razorpay payment webhook processing.
- Developed custom, dependency-free SVG analytics pipelines to visualize task completion matrices and habit consistency heatmaps in real-time.

WordSage — AI-Powered Enterprise Writing SaaS

Live Platform — [GitHub Repository](#)

- Architected and deployed a production-grade AI SaaS platform delivering domain-aware writing assistance across industry-specific modes using composite prompt orchestration.
- Engineered secure multi-tenant collaboration infrastructure featuring team workspaces, role-based access control (RBAC), style-guide enforcement, document versioning, and real-time collaboration workflows.
- Designed and implemented a monetization ecosystem including subscription billing, coin-based economy, Razorpay payment processing, and HMAC-secured webhook verification.
- Built a scalable backend architecture with 24 normalized PostgreSQL tables, atomic Prisma transactions, NextAuth JWT authentication, rate limiting, and layered security controls.
- Containerized the complete application stack using Docker Compose and Nginx reverse proxy, enabling production-ready multi-service deployment and health-aware orchestration.

EDUCATION

Bachelor of Science in Computer Science

MVLU, Mumbai

Oct 2022 – Aug 2025

